PRODUCT REQUIREMENTS DOCUMENT · V2.0 · COURSE FORMAT

MPSTME GUARDIAN

Collaborative Dual-UAV Perimeter Surveillance of the NMIMS Main Building, Vile Parle (West), Mumbai

MuJoCo · Problem-Based Learning · MDRIIA Group 05

> *"To what extent can a collaborative dual-UAV surveillance system simulated in MuJoCo optimize campus perimeter patrol cycle time and OpenCV human detection latency compared to static security guard patrols?"*

COURSE	MDRIIA — Modern Day Robotics & Its Industrial Applications (702CO0E012)
PROGRAMME	B.Tech CSBS, Semester VI, AY 2026-27 · SVKM's NMIMS MPSTME, Mumbai
REPOSITORY	`Group_05_Muaaz_Aditya_Zaid_Soumya/` — course scaffold format (the course faculty)
SCENE	NMIMS Main Building, V.L. Mehta Road × Gulmohar Road — OpenStreetMap geometry, official virtual-tour appearance
TECH LEAD	E043 Muaaz Mohammed Iqbal Shaikh (SAP 70362400030) — flight, fleet, integration, owner of `main`
MEMBERS	E051 Aditya Rajkumar · E075 Zaid Rezaur Rahman · E077 Soumya Subhankar Ranasingh
SUPERSEDES	PRD v1.0 (pre-scaffold prototype, invented 120 × 90 m campus)
DATE	8 October 2026

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1 Executive Summary

MPSTME GUARDIAN simulates two cooperating quadrotors patrolling the real perimeter of the NMIMS Main Building in Vile Parle (West), and measures — rather than asserts — how much faster they revisit every point on that perimeter, and how quickly they detect an intruder, than a guard on foot. The repository now follows the course scaffold exactly: one MJCF model, one control file with a TODO region per student, and an analytics folder whose benchmark CSV feeds the 4-page IEEE manuscript.

342 m	±0.03 m	41.0 s	0
REAL PATROL RING (OSM)	ALTITUDE HOLD IN GUSTS	WORST REVISIT, OFFSET RING	COLLISIONS, ALL RUNS

Everything in that row is measured in MuJoCo on the generated NMIMS scene. Three further results from this revision change what the team must build:

- **The guide's APF does not guarantee separation at cruise speed.** In a measured head-on pass at 6 m/s per aircraft, APF alone (k_{rep} = 8.5, d_{safe} = 2.5 m) let the centres reach **0.40 m** — contact. A layered scheme (altitude layers + closing-speed governor + right-hand sidestep, APF kept as the inner layer) holds **3.29 m** and the pair passes cleanly.
- **The level-camera pinhole formula is wrong by metres in flight.** Measured tilt during cruise is 7.2° median and 28.5° at p95, which displaces the projected ground point by 2.5–3.0 m median and **11–13 m at p95** — wider than the 5 m geofence warning buffer.
- **The scaffold's figure generator fabricates the benchmark CSV** from `np.random.normal`. It has been rewritten to read data only and to refuse any CSV that lacks a provenance record. The scaffold CSV is preserved untouched as a placeholder.

2 What Changed from v1.0

AREA	V1.0 (PROTOTYPE)	V2.0 (THIS DOCUMENT)
Repository format	Own layout: `src/patrol.py`, `run_patrol.py`, one file per owner	Course scaffold: `src/aerial_patrol_swarm.py` with per-student TODO regions; `analytics/`; `models/campus_perimeter_patrol.xml`
Campus	Invented 120 × 90 m arena, 380 m ring	Real NMIMS Main Building campus from OpenStreetMap; 342 m ring
Ownership unit	Whole files	Files *and* marked regions inside the shared control file, enforced by `src/check_region_ownership.py`
Branches	`feat/<roll>-<slug>`	Course-assigned: `feat/e043-lead-uav-flight-dyna`, `feat/e051-computer-vision-open`, `feat/e075-restricted-zone-geo-`, `feat/e077-csbs-campus-security`
Trial count	N = 25 per configuration	N ≥ 80 (guide §4), paired by seed across conditions
Separation	14 m vertical deconfliction	Guide APF retained, plus altitude layering, governor and sidestep — with measured evidence for why
Rotor layout	X-config, 0.22 m arms, 1.25 kg	"+" config, l = 0.26 m, m = 1.35 kg — matching the guide
Economics	Currency-based CapEx/OpEx	Strictly dimensionless (course rule: no currency symbols)

3 Simulation Scene: NMIMS Main Building

The simulation should be recognisably MPSTME's own institution. The NMIMS Main Building on V.L. Mehta Road was chosen over the MPSTME engineering block because NMIMS publishes an official virtual tour of it, which gives a photographic reference for appearance; the engineering block (OSM way 1056855968, 11 storeys, inside the SVKM compound on Gulmohar Road) remains available as a second scene using the same generator.



Figure 1 — Generated scene, overview camera. NMIMS Main Building (white, ten bands), boundary wall, trees on the V.L. Mehta Road and Gulmohar Road frontages, autos and tempos parked along the wall, traffic signals at the junction, and 42 surrounding buildings for context and occlusion. Green discs are the nine patrol checkpoints.

3.1 Sources and what each contributes

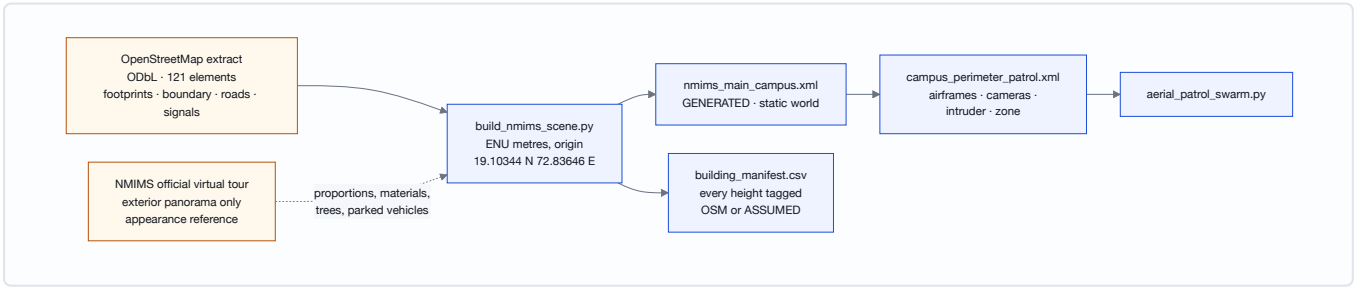


Figure 2 — Scene build. Geometry is measured (OSM); appearance is referenced (tour). No tour imagery is embedded in the simulation. The tour's 16 interior panoramas are not used — a perimeter UAV never sees them, and a 360° photograph carries no depth from which to recover 3-D geometry.

3.2 Measured versus assumed

ELEMENT	VALUE	SOURCE / STATUS
Main Building footprint	3,606 m²	OSM way 791070239 — MEASURED

ELEMENT	VALUE	SOURCE / STATUS
Campus boundary	6,473 m², 320 m	OSM way 1170351966 — **MEASURED**
Roads, junction signals	115 segments, 3 signals	OSM highways — **MEASURED**
Main Building height	10 storeys × 3.6 m = 36 m	Storeys counted from the tour panorama (top partly cropped) — **ASSUMED**
Surrounding building heights	By OSM type	Only a few carry `building:levels` in OSM — **ASSUMED**
Tree canopy tops	≈ 15 m	Judged from the panorama — **ASSUMED**; drives cruise altitude
Boundary wall	2.4 m	**ASSUMED**
Electrical substation zone	18 × 8 m, north-west yard	Location not known — **ASSUMED**; confirm with the estates office



Figure 3 — Validation against the real campus. Left: the official NMIMS virtual-tour exterior panorama (© NMIMS, reproduced for academic comparison). Right: the MuJoCo `tour_viewpoint` camera placed at the same junction. The real image is a panorama projection, so the comparison is qualitative: massing, facade rhythm, frontage trees and parked vehicles match; exact heights do not, and are flagged as assumed.

3.3 Patrol geometry

- **Ring:** *Superseded on 29 Oct 2026: all patrolling is inside the college. The ring is now the boundary-wall line offset 3.0 m **inward** (9 checkpoints, 298.6 m; models/scene_source/build_nmims_scene.py, PATROL_INSET; tests/test_e043_exterior_route_inside_campus.py).* The v2.0 design below this line (3 m outward, 342 m) is kept only as history.
- **Altitude:** Alpha 20 m, Bravo 24 m. The guide's 12 m would put both aircraft inside the ≈ 15 m tree canopy on two frontages; the 4 m layer separation is the strategic layer of §6.3.
- **Launch:** rooftop docks on the 36 m roof, transit at 44 m (clears the lift machine room), then descend to the patrol layer.

4 Problem & Baseline Status

The course guide and scaffold supply target figures. Under the zero-hallucination mandate they are **targets to be replaced by measurement**, never results. On the real NMIMS perimeter several of them will not hold, and that is expected.

FIGURE (GUIDE / SCAFFOLD)	VALUE	STATUS	RESOLUTION
Guard perimeter cycle T_{guard}	48.72 min	PLACEHOLDER	E077 — define the comparator (perimeter-only walk vs. full building round) and measure it. A 342 m perimeter walk at 1.2 m/s takes ≈ 5 min; 48 min only makes sense for a round that includes floors.
Dual-UAV cycle T_{UAV}	7.84 min	SUPERSEDED	Measured on the real ring: **0.69 min** worst-case revisit (offset ring, 3 smoke trials).
Speedup $S = T_{guard}/T_{UAV}$	6.21×	PLACEHOLDER	Recompute from both measured columns.
Labour reallocation η	0.70 (50.4 / 72 guard-h/day)	PLACEHOLDER	E077 — from the actual NMIMS security roster.
Normalised OpEx C_{hybrid}, CapEx K	0.66, 0.31	PLACEHOLDER	E077 — sourced assumptions or a parametric sensitivity band.
Payback P	10.94 months	PLACEHOLDER	Derived from the above; reported as a band.
Blind spots, CCTV → UAV	35.8 % → 1.2 %	PLACEHOLDER	E075 — computed from camera footprints with tree and building occlusion.
Vision latency, 1000 → 5 lx	28.42 → 49.24 ms	PLACEHOLDER	E051 — wall-clock timing of the real detector per illumination tier.
IoU across illumination	≥ 0.738	PLACEHOLDER	E051 — against projected ground-truth boxes.
`campus_patrol_benchmark.csv`	80 rows	SYNTHETIC	Generated by `np.random.seed(42)` in the scaffold. Replaced only by `run_benchmark.py`.

5 Objectives & Success Metrics

ID	OBJECTIVE	METRIC (CSV COLUMN)	TARGET	OWNER
01	Shorten the time any perimeter node is unobserved	`uav_cycle_min` vs `guard_cycle_min`	Significant, paired by seed	**E043 / E077**
02	Shorten time-to-detect an intrusion	`uav_detect_lat_s` vs `guard_detect_lat_s`	Significant; randomised appearance	**E051 / E077**
03	Show the coordination policy drives the gain	offset ring vs sectors, equal energy	CI-separated	**E043**
04	Never lose separation	`min_sep_m`	≥ 2.5 m in every trial	**E043**
05	Real-time detection across illumination	latency (ms), IoU per lux tier	< 50 ms (20 FPS); IoU > 0.70	**E051**
06	Correct, stable zone classification	4-tier FSM accuracy, chatter count	No chatter at the boundary	**E075**
07	Close perimeter blind spots	`blind_spot_pct`	Reported with occlusion	**E075**

ID	OBJECTIVE	METRIC (CSV COLUMN)	TARGET	OWNER
08	Dimensionless operational case	S, η, ΔOpEx, P	Band, from measured inputs	**E077**

6 System Architecture

6.1 Components and ownership in the course layout

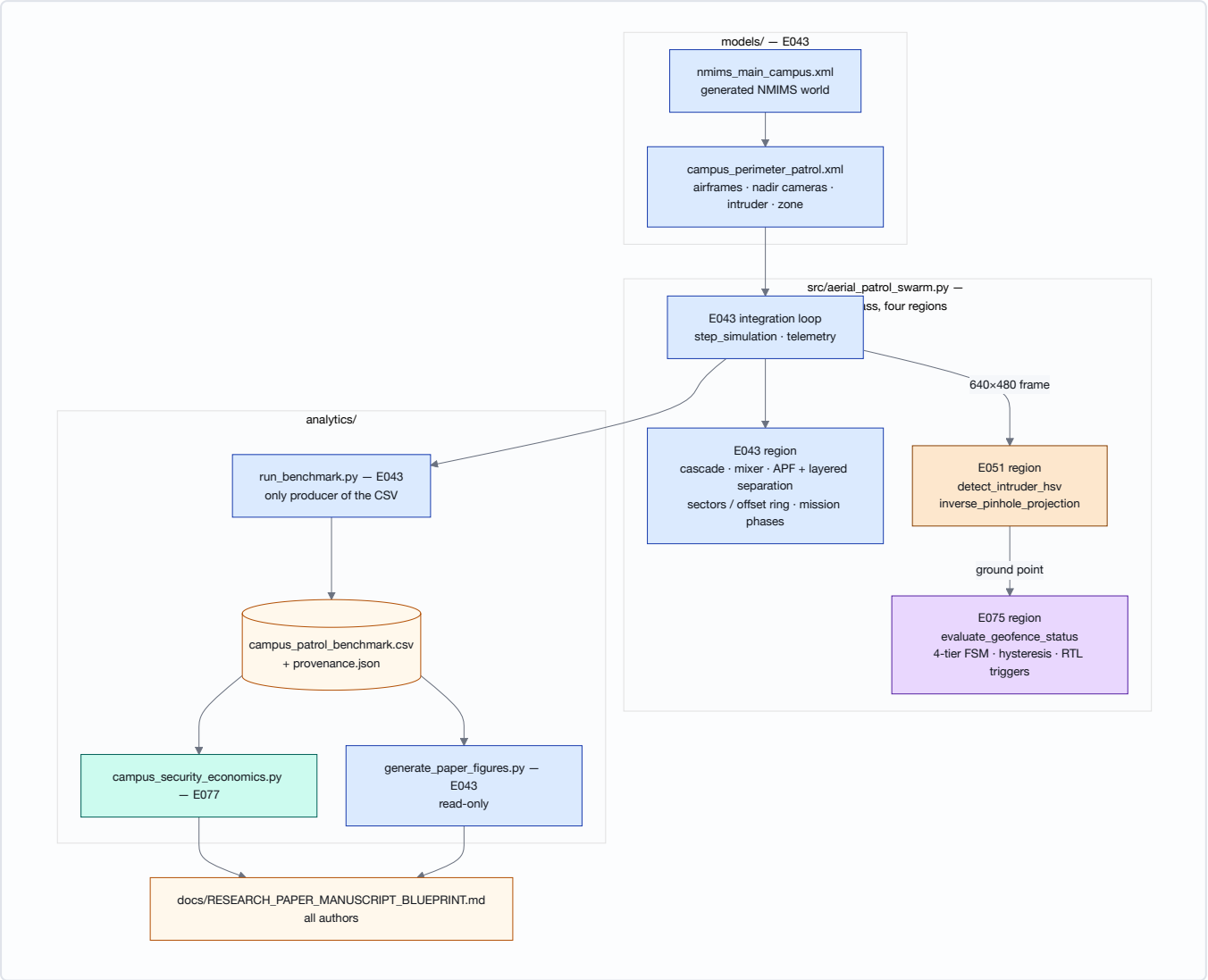


Figure 4 — Components coloured by owner. Blue E043 · orange E051 · purple E075 · teal E077. The shared control file is split by `>>> E0xx` / `<<< E0xx` markers; a pull request may change only lines inside its author's region.

6.2 Flight control

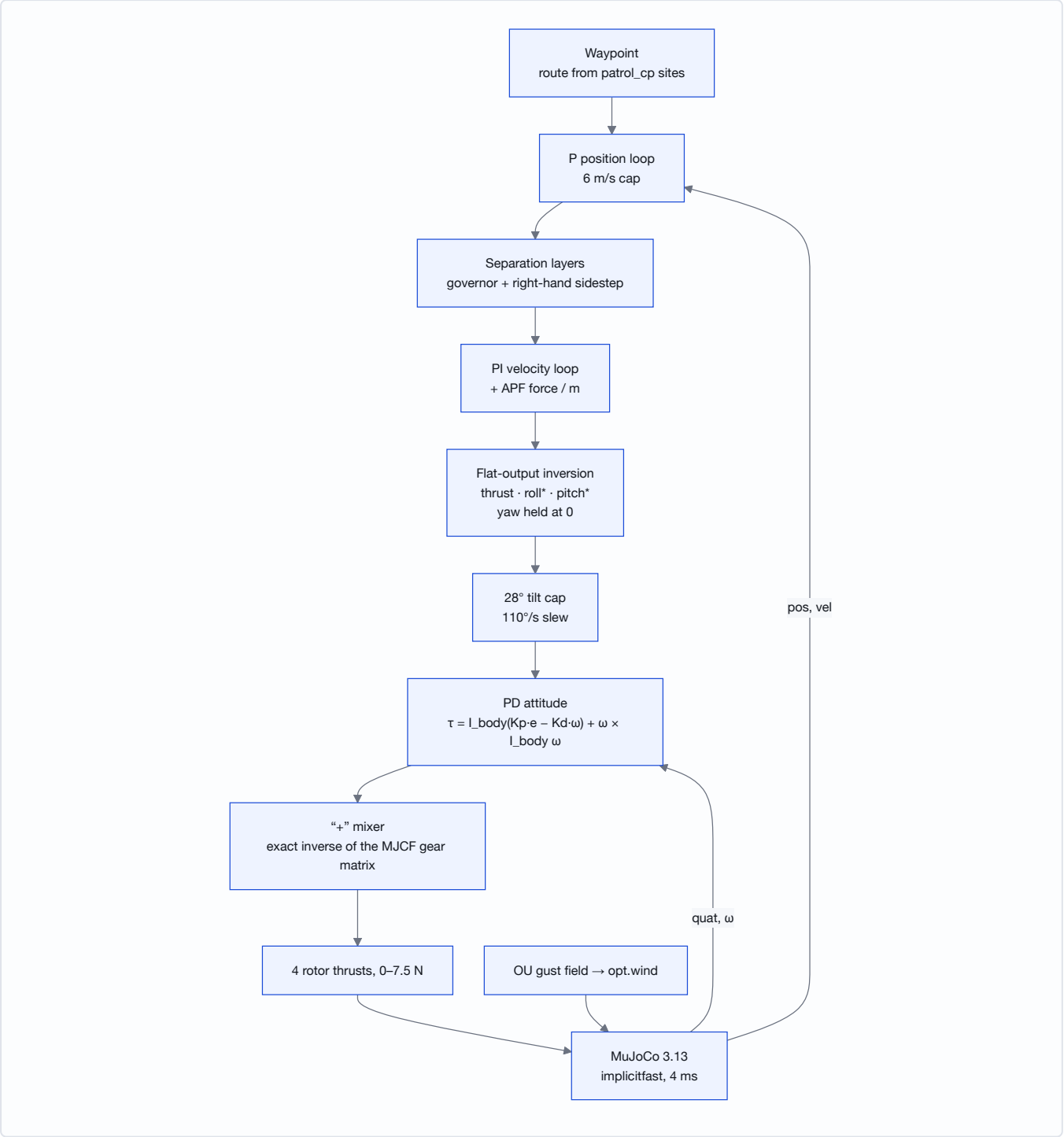


Figure 5 — Cascade. Two details that are easy to get wrong and will be probed at the viva: MuJoCo reports inertia in the body's *principal* frame (here reordered so I_{zz} is listed first), so it is rotated back to body axes before use; and yaw is held at zero so the nadir image axes stay aligned with East/North, which the pinhole projection relies on.

ALLOCATION FOR THE "+" LAYOUT

```
τx = l (F2 − F3)        τy = l (F1 − F0)
τz = kQ (F0 + F1 − F2 − F3)        T = F0 + F1 + F2 + F3

guide notation:  τB = [ l(T4−T2),  l(T1−T3),  cQ(T1−T2+T3−T4) ]
                 with T1 = rotor1, T2 = rotor3, T3 = rotor0, T4 = rotor2
```

6.3 Separation: why three layers

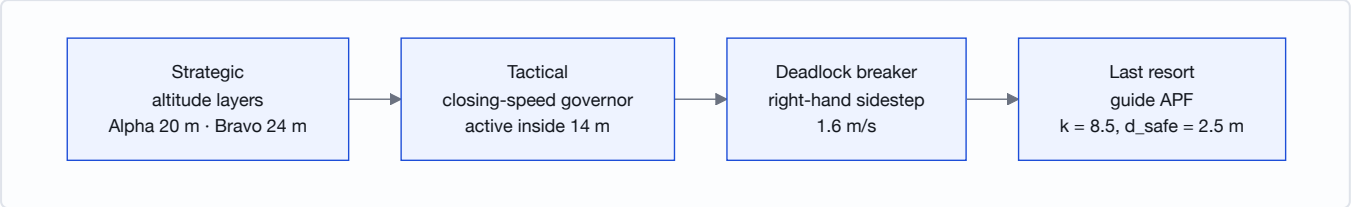


Figure 6 — Layered separation. Evidence (`tests/test_e043_separation.py` , head-on at 6 m/s each, layers disabled): APF alone reaches 0.40 m — contact; with governor and sidestep, 3.29 m and the pair passes. Analytically, the work APF can do from d_{safe} to d is $(k/2)(1/d - 1/d_{safe})^2$; equating it to the 48.6 J closing energy gives d = 0.26 m.

6.4 Mission and alert states

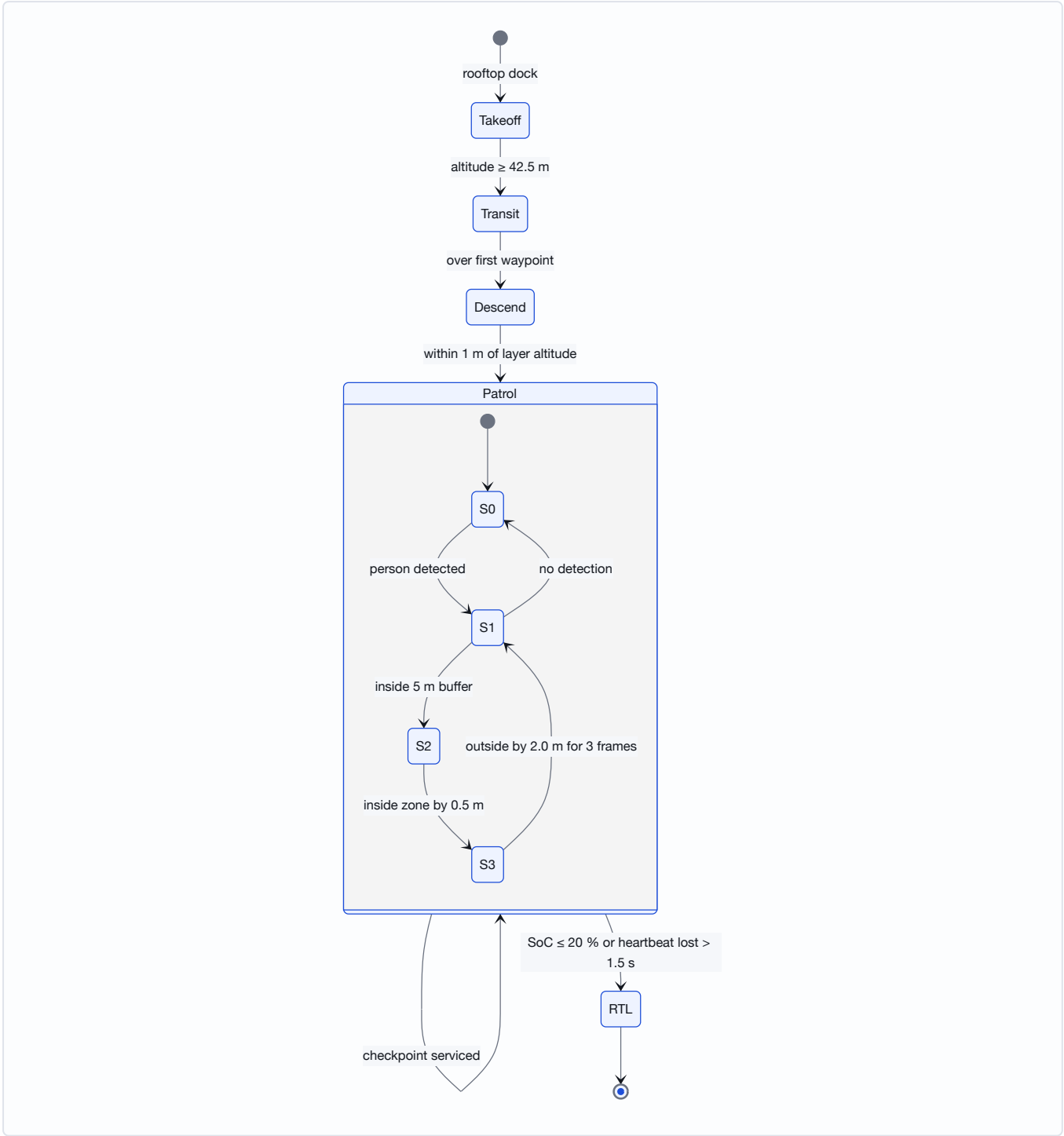


Figure 7 — Flight phases (E043, built) and the guide's 4-tier alert FSM (E075, to build). Hysteresis thresholds and RTL triggers are taken from the guide's viva answers. RTL is triggered by E075's FSM and executed by E043's flight layer.

6.5 Detection and alert path

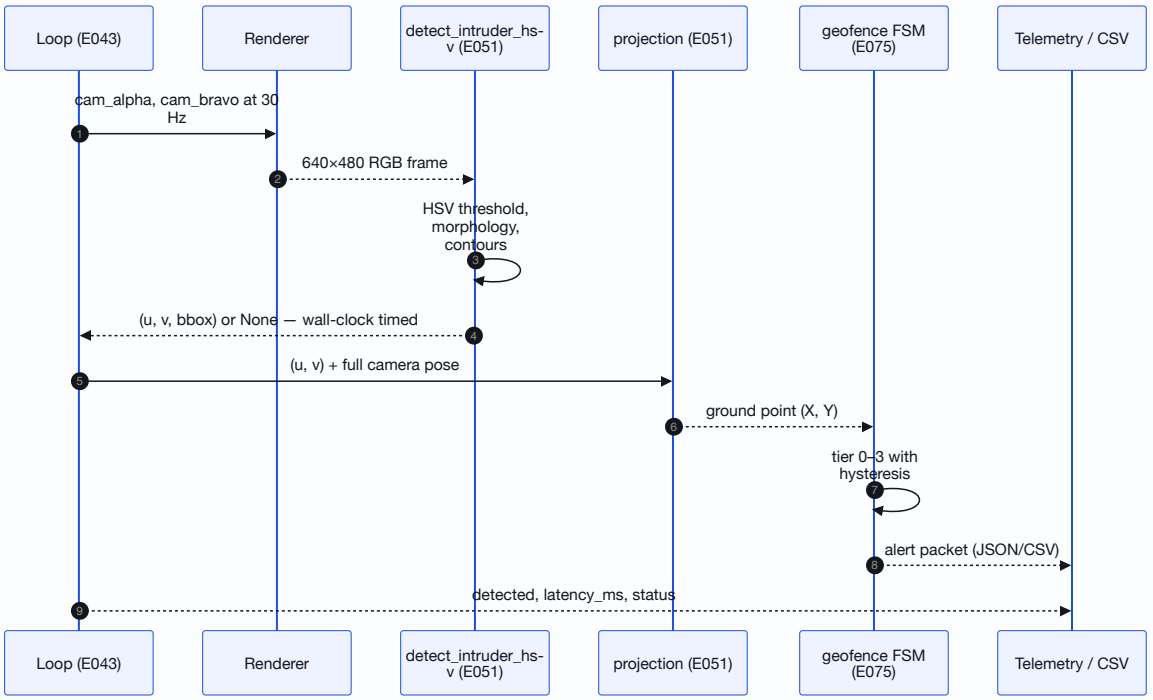


Figure 8 — Detection path. Until `detect_intruder_hsv` exists it raises `NotImplementedError`; the loop then records "not implemented" and reports **no** detections. The scaffold instead logged `28.5 + N(0, 2)` ms whenever the aircraft came within 12 m — synthetic data, now removed.

6.6 Evidence pipeline

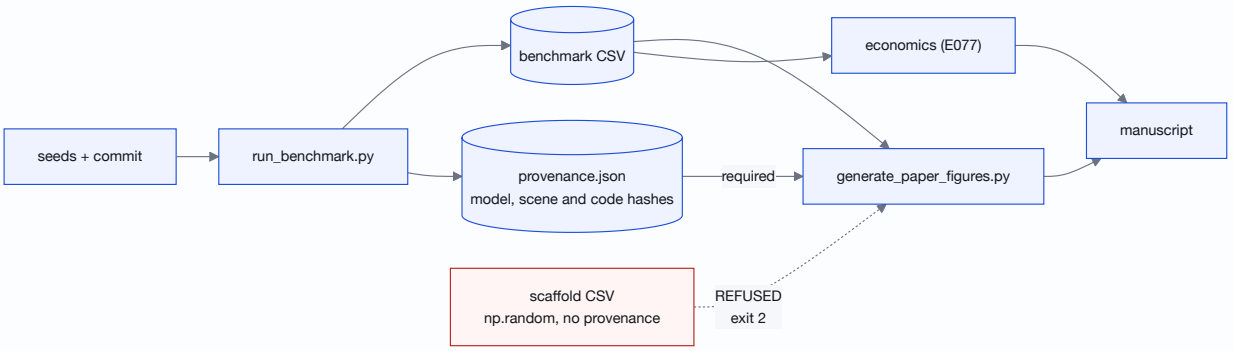


Figure 9 — No path from invented numbers to the paper. The figure script exits with code 2 on any benchmark CSV without a provenance record; `--allow-placeholder` exists only for layout previews and watermarks every figure "PLACEHOLDER DATA — NOT FOR SUBMISSION".

7 Scaffold Audit — Findings & Resolutions

The course scaffold is a starting point with deliberate student boundaries. Working through it on the real scene surfaced the issues below. Each was checked empirically before being recorded; the original scaffold is preserved untouched in `_scaffold_original/` for comparison.

#	FINDING	EVIDENCE	RESOLUTION / OWNER
1	Nadir cameras point at the sky	`euler="3.1415 0 0"` flips the camera's –z view axis to +z; rendered frame is entirely black	Rotation removed — **FIXED** E043
2	No yaw authority	All rotor gears `0 0 1 0 0 0` — no drag torque, so no yaw moment can be produced	tz = ±0.0201 per rotor pair — **FIXED** E043
3	Arm length mismatch	Guide l = 0.26 m; MJCF sites at 0.20 m	Aligned to 0.26 m — **FIXED** E043
4	Focal length mismatch	fovy 60° → f = 415.7 px; guide uses f = 450 px	fovy 56.14° → 450 px — **FIXED** E043
5	Figure script fabricates the benchmark CSV	`np.random.normal(loc=48.72 …)`, overwrites on every run	Read-only + provenance gate — **FIXED** E043
6	Synthetic detection latency in the loop	`28.5 + np.random.normal(0, 2.0)` when within 12 m	Removed; latency is now wall-clock measured — **FIXED** E043
7	Kinematic "flight"	`step_simulation` placed drones on a circle; no physics	Full MuJoCo closed loop — **FIXED** E043
8	APF separation claim	Head-on at cruise: 0.40 m, contact	Layered separation — **FIXED** E043
9	Level-camera projection	Ground error 2.5–3.0 m median, 11–13 m p95 at measured tilt	Use full camera pose — **OPEN** E051
10	Geofence bounds hard-coded to the old scene	`(13, 15, 23, 25)` no longer matches any zone	Use `self.substation_bounds` read from the model — **OPEN** E075
11	Economics constants hard-coded	48.72, 7.84, 0.66, 0.31 typed into the model	Read inputs from the benchmark CSV — **OPEN** E077
12	Blueprint trial count	Abstract and §V say N = 50; guide §4 requires ≥ 80	Use N ≥ 80 — **OPEN** all
13	Related Work text does not match the papers	e.g. [5] Mittal is an object-detection survey but is described as collision avoidance; [3] Wu is task allocation, described as mechanical damping. All six DOIs verified genuine against Crossref.	Rewrite §II from the papers themselves — **OPEN** all
14	Pre-written conclusions	Blueprint states p < 0.001 "across all metrics" before any data exists	Write results last, from the CSV — **OPEN** all
15	Red intruder avatar	Makes HSV detection trivially easy; real people are not red	Report as a limitation; the scene's red tempo cabs and signal lamps are a built-in false-positive test — E051

8 Functional Requirements

ID	REQUIREMENT	OWNER	VERIFICATION / STATUS
FR-01	The scene shall reproduce the NMIMS Main Building campus from measured footprints, with every assumed value listed in a manifest.	E043	Generator + manifest MET
FR-02	All scaffold element names shall be preserved so `src/test_env.py` passes unmodified.	E043	test_env passes MET
FR-03	Each airframe shall be actuated only by four rotor thrusts, with yaw from rotor drag torque.	E043	Mixer is the exact inverse of the gear matrix MET
FR-04	Altitude shall be held within ±0.5 m in gusts.	E043	±0.033 m measured MET
FR-05	Inter-UAV separation shall stay ≥ 2.5 m in every trial, including head-on encounters.	E043	3.29 m head-on; ≥ 6.3 m in patrol MET
FR-06	Both the guide's sector policy and an offset-ring policy shall be selectable.	E043	`--policy` MET
FR-07	The benchmark CSV shall be produced only by seeded simulation, with provenance.	E043	`run_benchmark.py` MET
FR-08	Intruder appearance time shall be randomised per seed.	E043	Uniform 40–160 s MET
FR-09	An RTL behaviour shall return the aircraft to its dock when E075 raises the trigger.	E043	OPEN
FR-10	The scene shall offer four illumination tiers (1000, 250, 50, 5 lx equivalents).	E043	OPEN
FR-11	Detection shall use only the rendered RGB frame — never simulator state.	E051	OPEN
FR-12	Ground projection shall use the full camera pose (tilt and yaw), not the level-camera formula.	E051	Error budget: < 1 m p95 OPEN
FR-13	IoU shall be scored against ground-truth boxes projected from the intruder geometry.	E051	OPEN
FR-14	Latency and IoU shall be reported per illumination tier in `vision_illumination_benchmark.csv`.	E051	OPEN
FR-15	The geofence shall read zone geometry from the model and test true polygons by ray casting.	E075	OPEN
FR-16	The 4-tier FSM shall apply hysteresis (enter 0.5 m inside, clear 2.0 m outside for 3 frames).	E075	OPEN
FR-17	RTL shall trigger at SoC ≤ 20 % or heartbeat loss > 1.5 s.	E075	OPEN
FR-18	Alert packets shall be serialised to JSON/CSV with tier, time, aircraft and ground point.	E075	OPEN
FR-19	`blind_spot_pct` shall be computed from camera footprints with tree and building occlusion.	E075	OPEN
FR-20	The guard comparator shall be defined and measured, filling `guard_cycle_min` and `guard_detect_lat_s`.	E077	OPEN
FR-21	The economics model shall read its inputs from the benchmark CSV and contain no currency symbols.	E077	OPEN

ID	REQUIREMENT	OWNER	VERIFICATION / STATUS
FR-22	Every pull request shall pass `check_region_ownership.py` before merge.	E043	Gate built and self-tested MET

9 Non-Functional Requirements

ID	ATTRIBUTE	REQUIREMENT
NFR-01	Reproducibility	A seed reproduces its trial; provenance records model, scene and controller hashes.
NFR-02	Runtime	Measured: 150 s simulated in 5.8 s with vision off. The 80-trial batch must fit in one sitting even with vision on.
NFR-03	Portability	`pip install mujoco numpy scipy matplotlib opencv-python`; no GPU required.
NFR-04	Attribution	Every line in the shared file belongs to exactly one roll number (region markers).
NFR-05	Honesty of stand-ins	Unimplemented parts raise `NotImplementedError` or write NaN — never a plausible default.
NFR-06	Format fidelity	The course directory tree is preserved; additions are listed in the README tree.

10 Data Specification

ANALYTICS/CAMPUS_PATROL_BENCHMARK.CSV — COURSE SCHEMA, UNCHANGED

COLUMN	OWNER	DEFINITION
`trial_id`	E043	1…N; each maps to a seed in the provenance record
`uav_cycle_min`	E043	Worst-case gap between consecutive services of any checkpoint, both aircraft merged, in minutes
`min_sep_m`	E043	Minimum 3-D inter-UAV distance while both are on patrol
`uav_detect_lat_s`	E051	First detection time minus intruder appearance time (simulated seconds)
`blind_spot_pct`	E075	Share of the perimeter wall line never inside an unoccluded camera footprint during the trial
`guard_cycle_min`	E077	Guard comparator cycle for the same perimeter, under the stated definition
`guard_detect_lat_s`	E077	Guard time-to-detect for the same intrusion schedule

Empty cells mean “owner’s code not yet merged”. They are never filled with a constant.

OTHER ARTEFACTS

- `campus_patrol_benchmark.provenance.json` — UTC time, commit, policy, trials, wind range, model/scene/controller hashes, Python and MuJoCo versions, per-trial seed, wind and appearance time.
- `analytics/runs/telemetry_*.csv` — per-run flight telemetry at 30 Hz (gitignored; regenerable from seed).
- `analytics/vision_illumination_benchmark.csv` (E051) — `lux, mean_latency_ms, std_latency_ms, iou_median, iou_iqr, n_frames`.

11 Experimental Design

FACTOR	LEVELS	WHY
Coordination policy	sectors (guide) · offset ring	Isolates the policy effect — the paper's contribution
Seed	≥ 80, shared across policies	Pairs trials, so a paired t-test is legitimate (blueprint specifies paired)
Wind	mean 0–6 m/s, OU gusts	Rejects the "calm air only" criticism
Intrusion time	uniform 40–160 s per seed	Latency must measure the system, not one lucky geometry
Illumination	1000 · 250 · 50 · 5 lx tiers	Guide's latency/IoU-versus-light analysis

Statistics: paired two-tailed t-test per metric, with mean difference, 95 % CI and effect size; IoU reported as median and IQR; every plotted mean carries a standard-error bar and its n.

12 CSBS Dimensionless Economics

The course forbids currency. The guide's structure is kept; its inputs change from typed constants to measurements and sourced ratios:

```
S_patrol   = guard_cycle_min / uav_cycle_min                    (both from the CSV)
eta_labor  = H_reallocated / H_total                            (NMIMS roster; E077)
dOpEx      = 1 − C_hybrid / C_guard-only     with C_guard-only ≡ 1.00
P_payback  = K_CapEx,normalised / dOpEx      reported across a band of C_hybrid and K
```

THE COMPARATOR DECIDES THE HEADLINE

A guard walking the 342 m perimeter needs about 5 minutes; a guard completing a full round of a ten-storey building needs far longer. S_{patrol} can be 7× or 70× depending on which is chosen. E077 must fix the definition first, in writing, and the paper must state it.

13 Risks

ID	RISK	OWNER	MITIGATION
R-01	Scaffold placeholder numbers reach the manuscript	E043	Figure gate; review checklist; §4 status table at every merge
R-02	Detector not finished in time	E051	Starts day 1 with offline frame fixtures; loop already degrades honestly
R-03	Merge conflicts in the shared control file	E043	Region markers + ownership gate; daily integration
R-04	Assumed heights or zone location are wrong	E077 / E075	Site check; one-line change in the generator, then regenerate
R-05	Geofence misfires from projection error	E051 / E075	FR-12 pose-aware projection; hysteresis
R-06	A member cannot defend their region at the viva	All	Nobody writes inside another member's region

14 Integrity, Licensing, Privacy

- **COPE.** No number reaches the paper unless `run_benchmark.py` produced it with provenance.
- **OpenStreetMap.** Geometry is ODbL 1.0; the attribution “© OpenStreetMap contributors” must appear in the paper and the README.
- **NMIMS imagery.** The tour panorama is NMIMS’s photograph. It is used only for the side-by-side validation figure, with credit; confirm with faculty before including it in the public LinkedIn video.
- **Privacy.** Detection is person-presence only — no identity, no faces, no biometric storage.
- **Deployment.** Simulation only. Any physical flight over Vile Parle would fall under Indian civil UAS rules and the airport-vicinity airspace map; nothing here is a compliance determination.

15 Milestones

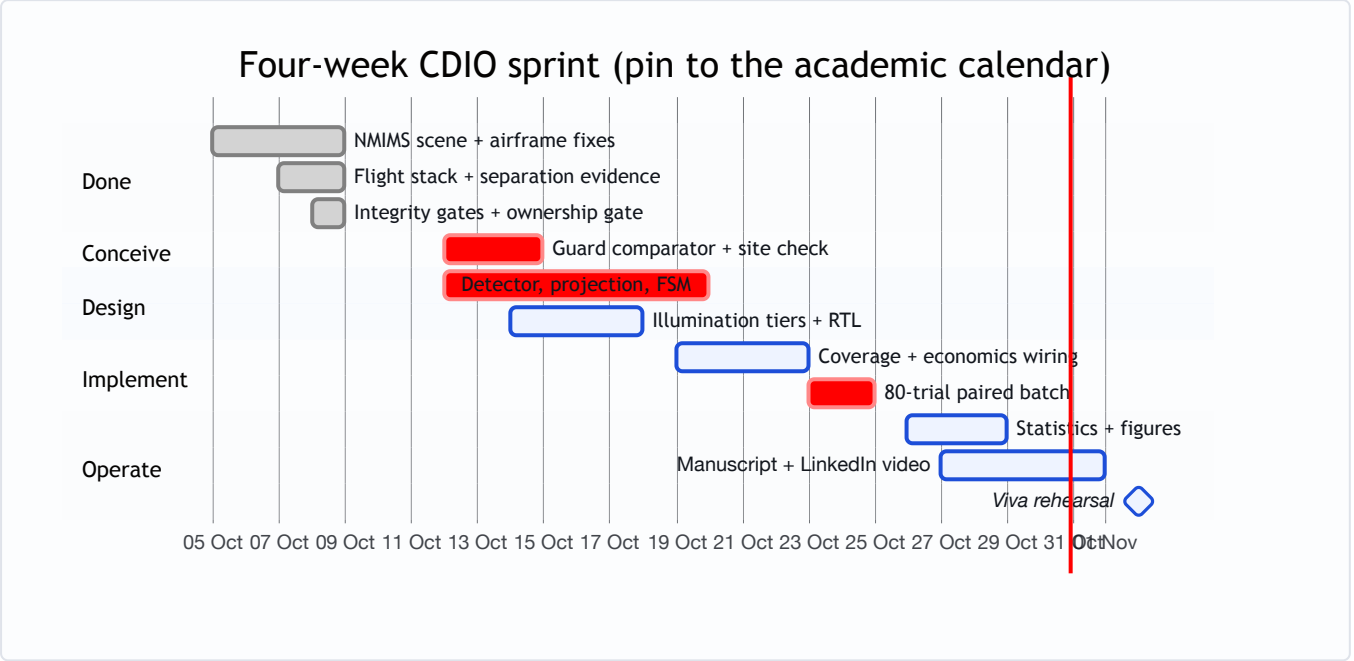


Figure 10 — Milestones. Dates are indicative from 12 October 2026.

16 Open Questions

ID	QUESTION	OWNER
Q-01	Does the course faculty accept the NMIMS Main Building scene in place of the scaffold's generic 60 × 60 m campus?	E043
Q-02	Are the scaffold fixes in §7 (camera orientation, yaw torque, figure script) acceptable changes to "provided" files?	E043
Q-03	Guard comparator: perimeter-only walk, or the full security round including floors?	E077
Q-04	Where is the electrical substation on the NMIMS Main campus, and how many storeys does the building have?	E075 / E077
Q-05	May the tour panorama appear in the public LinkedIn video?	E043

PRD v2.0 · companion: *Implementation & Review Plan v2.0*. Contains information from OpenStreetMap, © OpenStreetMap contributors, ODbL 1.0.